

Answer **all** questions in the spaces provided.

1 A student investigated mitosis in the tissue from an onion root tip.

0 1 . **1** The student prepared a temporary mount of the onion tissue on a glass slide. She covered the tissue with a cover slip. She was then given the following instruction.

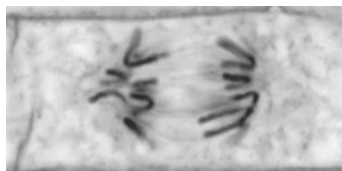
“Push down hard on the cover slip, but do not push the cover slip sideways.”

Explain why she was given this instruction.

[2 marks]

Figure 1 shows one cell the student saw in the onion tissue.

Figure 1



0 1 . **2** The student concluded that the cell in **Figure 1** was in the anaphase stage of mitosis. Was she correct? Give **two** reasons for your answer.

[2 marks]

1

2

0 1 . 3 The student counted the number of cells she observed in each stage of mitosis. Of the 200 cells she counted, only six were in anaphase.

One cell cycle of onion root tissue takes 16 hours. Calculate how many minutes these cells spend in anaphase.

Show your working.

[2 marks]

Answer = _____ minutes

Turn over for the next question